



Department of Information Technology
Academic Year 2023 - 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / Equivalence of pushdown automata and CFG-CFG to PDA
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO9
- **Activity Chosen:** Reflection
- **Justification:**

In unit III, Equivalence of pushdown automata and CFG-CFG to PDA is an important topic. It is difficult to remember the conversion from CFG to PDA and PDA TO CFG. This activity helps the students to recall the different conversion.

- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**
 - I explained the topic Equivalence of pushdown automata and CFG-CFG to PDA in the previous days.
 - The time allotted for the topic is 5 Minutes. The students during the activity as shown in Figure 1,2
 - The students were asked to study the concepts of PDA to CFG,CFG to PDA
 - Next day asked the students write their doubts in a paper.
 - The papers were collected and corrected as shown in Figure 3

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO3	3	3	3	1	2	1	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	3	3	1	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of equivalence of pushdown automata and CFG-CFG to PDA	The students will be able to identify the need for equivalence of pushdown automata and CFG-CFG to PDA of complex engineering problems	Able to design solutions for any applications using equivalence of pushdown automata and CFG-CFG to PDA	Able to analyze and apply equivalence of pushdown automata and CFG-CFG to PDA concept	Able to follow ethics for solving the problem	Able to implement the problem individually	Able to implement equivalence of pushdown automata and CFG-CFG to PDA.	Ability to recognize the need for equivalence of pushdown automata and CFG-CFG to PDA in real world applications	Exhibit knowledge in equivalence of pushdown automata and CFG-CFG to PDA for application development	Exhibit knowledge in equivalence of pushdown automata and CFG-CFG to PDA in TOC of IoT, Cloud Computing and Data Analytics to provide IT solutions	Exhibit knowledge in equivalence of pushdown automata and CFG-CFG to PDA in software projects to deliver a quality product.

- Images / Screenshot of the practice:



Figure 1: Reflection



Figure 2: Reflection

Ashmitha KV

CS3402 - Theory of Computations
Reflection Activity

Equivalence of Pushdown Automata and CFG + CFG to PDA:

CFG to PDA:
This is a language in which equal number of a's followed by equal number of b's ($L = \{a^n b^n / n \geq 1\}$). The logic for this PDA can be applied as follows we will push all a's onto stack

Design:

$$\begin{aligned} \delta(q_0, a, z_0) &= (q_0, a z_0) \\ \delta(q_0, a, a) &= (q_0, aa) \\ \delta(q_0, b, a) &= (q_1, \epsilon) \\ \delta(q_0, b, a) &= (q_1, \epsilon) \\ \delta(q_1, \epsilon, z_0) &= (q_2, \epsilon) \end{aligned}$$

where q_0 is a start state q_2 is accept state

Simulation:

$$\begin{aligned} (q_0, aaa, bbb, z_0) &\vdash (q_0, aaab, az_0) \\ &\vdash (q_0, abbb, aaz_0) \\ &\vdash (q_0, bbb, aaaz_0) \\ &\vdash (q_1, bb, aaaz_0) \\ &\vdash (q_1, bb, aaaz_0) \\ &\vdash (q_1, b, aaz_0) \\ &\vdash (q_1, \epsilon, z_0) \\ &\vdash (q_2, \epsilon) \end{aligned}$$

Accept state

G. Kavipraba
953622205084
B.Tech - IT

Activity - reflection
Topic - Normal forms, simplification of CFG
Equivalence of pushdown automata and CFG - CFG to PDA

(i) Normal form:

There is two normal forms
↳ Chomsky's forms
↳ Greibach Normal form

Chomsky's Normal form:

$$\begin{aligned} \text{Non-terminal} &\rightarrow \text{Non-terminal} \cdot \text{Non-terminal} \\ \text{Non-terminal} &\rightarrow \text{terminal} \end{aligned}$$

Ex:

$$\begin{aligned} S &\rightarrow AP_1 \\ P_1 &\rightarrow AP_2 \\ P_2 &\rightarrow AP_3 \\ A &\rightarrow a \end{aligned}$$

Greibach Normal form:

$$\begin{aligned} \text{Non-terminal} &\rightarrow \text{terminal} \cdot \text{Non-terminal} \\ \text{Non-terminal} &\rightarrow \text{terminal} \end{aligned}$$

Ex:

$$\begin{aligned} S &\rightarrow aAA & \text{GNF} \\ S &\rightarrow aAX \\ X &\rightarrow a \end{aligned}$$

S Aarthi
953622205002

Reflection

Equivalence of pushdown Automata and CFG

For converting CFG to push down Automata, the production rule

$$\begin{aligned} \text{Non-terminal} &\rightarrow \text{terminal (list of non-terminals)} \\ \text{Non-terminal} &\rightarrow \text{non-terminal Non-terminal (terminal)} \end{aligned}$$

Construct PDA

$$\begin{aligned} S &\rightarrow AB & A &\rightarrow CD \\ B &\rightarrow b & C &\rightarrow a \\ D &\rightarrow a \end{aligned}$$

Soln: First convert into Greibach Normal Form:

$$\begin{aligned} S &\rightarrow CDB & S &\rightarrow aDB \\ A &\rightarrow CD & A &\rightarrow aD \\ B &\rightarrow b & B &\rightarrow b \\ C &\rightarrow a & C &\rightarrow a \\ D &\rightarrow a & D &\rightarrow a \end{aligned}$$

Equivalent PDA:

$$\begin{aligned} \delta(q_1, a, S) &\rightarrow (q_1, DB) \\ \delta(q_1, a, A) &\rightarrow (q_1, D) \\ \delta(q_1, b, B) &\rightarrow (q_1, \epsilon) \\ \delta(q_1, a, C) &\rightarrow (q_1, \epsilon) \\ \delta(q_1, a, D) &\rightarrow (q_1, \epsilon) \end{aligned}$$

Simulate the string 'aab'

$$\begin{aligned} \delta(q_1, aab, S) &\vdash \delta(q_1, ab, DB) \\ &\vdash \delta(q_1, b, B) \\ &\vdash \delta(q_1, \epsilon, z_0) \\ &\vdash \delta(q_1, \epsilon) \end{aligned}$$

Accept the string.

Figure 3: Student Response Ms. K. Asmitha, G. Kavipraba and S.Aarthi

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students stated that they were able to quickly recall and remember the ideas of CFG-CFG to PDA and the equivalency of pushdown automata.
 - The students indicated that the activity helped them remember what was covered in the previous class and that it enhanced their ability to learn on their own.
 - It helps students learn about PDA to CFG and CFG to PDA conversion.

- ❖ ***Benefit of the practice:***

- Students can able to recall the concept of CFG-CFG to PDA and the equivalency of pushdown automata and it improves the problem-solving skills.
 - The students can share their doubts with neighbors.

- ❖ ***Challenges faced in implementation:***

- Some questions were difficult for students to understand the conversion.
 - I felt the slow learners were not understood clearly. I explained separately for that students, then they understood clearly.

References:

- ❖ <https://ohioline.osu.edu/>

- ❖ <https://www.nureva.com/blog/education/15-active-learning-activities-to-energize-your-next-college-class>

- ❖ <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member

HOD